

$$b^y = x \rightarrow \log_b x = y$$

Name: [REDACTED]Unit 6 Exponential & Logarithmic Functions TestTotal: **28** /30

PART 1: NO CALCULATOR

Circle the best response for questions #1 to #3.1. The equation of the vertical asymptote for the function $y = -2 \log_3(3x + 3) + 4$ is

☒ a) $x = -1$

b) $x = -2$

c) $x = 3$

d) $x = 4$

2. Solve. $2^x = 17^4$

a) $\log_2 17^4$

b) $2^{\frac{\log 2}{\log 17}}$

c) $\log_2 17 \times \log_2 4$

☒ d) $2^{\frac{4 \log 17}{\log 2}}$

$$= 3(x+1)$$

$$x = \log_2(17^4)$$

$$x = 4 \log_2 17$$

3. The equation $2 = \log_{(x+1)}(y+1)$ can be written as:

a) $y = \frac{2}{\log_{(x+1)} 1} - 2$

☒ b) $y = (x+1)^2 - 1$

c) $y = 2(x+1) - 1$

d) $y = \log_{(x+1)} 2 - 1$

4. Calculate the following, showing all steps required to solve without a calculator:

a) $\log_5 \sqrt[3]{25} = y$

b) $\log_4 8^2$

c) $8^{\log_2 4} = x$

$$5^y = 25^{\frac{1}{3}}$$

$$5^y = 5^{\frac{2}{3}}$$

$$y = \boxed{\frac{2}{3}}$$

$$(2^2)^y = (2^3)^2$$

$$2^{2y} = 2^6$$

$$2y = 6$$

$$y = \boxed{3}$$

$$(2) \log_8 x = \log_2 4$$

$$\log_2 8^y = \log_2 4$$

$$8^y = 4$$

$$2^{3y} = 2^2$$

$$3y = 2$$

$$y = \frac{2}{3}$$

$$x = \boxed{64}$$

(2)

b) Solve for x. Show all your work.

☒ a) $\log_{\frac{1}{10}} 10^x = -4$

b) $\log_x 6 = \frac{1}{2}$

$$\log_{\frac{1}{10}} (10)^x = -4$$

$$x \log_{\frac{1}{10}} (10) = -4$$

$$\left(\frac{1}{10}\right)^x = (10)^{-4}$$

$$(10^{-1})^x = 10^{-4}$$

$$-x = -4$$

$$x(-1) = -4$$

$$x = \boxed{4}$$

$$\left(\frac{1}{10}\right)^{-4} = 10^4$$

$$\left(\frac{1}{10}\right)^{-4} = 10^4$$

$$10^4 = 10^4$$

$$\log_x 6 = \frac{1}{2}$$

$$\log_x 6 = \frac{1}{2}$$

$$\left(x^{\frac{1}{2}}\right)^2 = (6)^2$$

$$x = 6^2$$

$$x = \boxed{36}$$

(2)

(2)

PART 1: NO CALCULATOR

- c) a) Determine at least two key points and graph the following function.

$f(x) = 3 \log \left[\frac{-1}{2}(x-4) \right] - 3$. **SHOW ALL WORK.** Label the function, and/or any asymptote(s) on the graph for full marks.

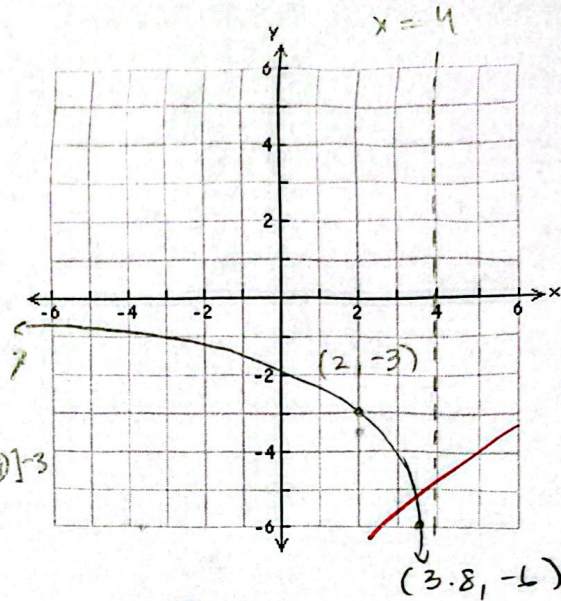
$\log_{10}$	
x	y
0	undef.
$\frac{1}{10}$	-1
1	0
10	1

$-2x+4$	$3y-3$
4	undef.
3.8	-6
2	-3
-16	0

key points: (3.8, -6) (2, -3)

$$\begin{aligned} -2(0)+4 &= 3(-1)-3 \\ -2(0.1)+4 &= 3(0)-3 \\ -2(1)+4 &= 3(1)-3 \\ -2(10)+4 &= 3(10)-3 \end{aligned}$$

$$f(x) = 3 \log_{10} \left[\frac{-1}{2}(x-4) \right] - 3$$



- b) Determine the following:

i) Domain

ii) Range

$$\{x \in \mathbb{R} \mid x < 4\}$$

$$\{y \in \mathbb{R}\}$$

- d) Solve for x, if $10 \log_{\sqrt{3}} 81^{3x} = 480$, show all your work.

$$10 \log_{\sqrt{3}} (81^{3x}) = 480$$

$$10 \log_{3^{\frac{1}{2}}} (3^4)^{3x} = 480$$

$$10(4)(3x) \log_{3^{\frac{1}{2}}} (3) = 480$$

$$34 = 3^{\frac{1}{2}}$$

$$120x \left(\frac{1}{2} \right) = 480$$

$$\frac{60x}{60} = \frac{480}{60}$$

$$x = 8$$

$$10 \left(3^{\frac{1}{2}} \right)^4 = 3^{24}$$

$$4 = 12$$

$$12 \times 40 = 480$$

$$10 \log_{\sqrt{3}} (81)^{38} = 480$$



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$$\log_b x = y \iff b^y = x$$

PART 2: CALCULATOR ALLOWED

Name: Crystal Ge

- e) Describe three characteristics of the function $g(x) = \log_{10} x$ that remain unchanged under the following transformation: a vertical stretch by a factor of 4 and a vertical translation of 2 units up

$$\log_{10} x \rightarrow 4 \log_{10} x + 2$$

x	y
0	undef.
$\frac{1}{10}$	-1
1	0
10	1

$$\rightarrow 4y + 2$$



① domain: the domain stays the same $\{x \in \mathbb{R} \mid x > 0\}$

② the VA: VA @ $x=0$ stays the same

③ no y-int: there is still no y-int

(3)

FORMULAE: $\text{pH} = -\log[\text{H}^+]$

$$\beta_2 - \beta_1 = 10 \log\left(\frac{I_2}{I_1}\right)$$

$$M = \log\left(\frac{1}{I_0}\right)$$

- f) Milk has a pH value of 6.4. How many times more acidic is orange juice, with a pH value of 4.0?

milk

$$\frac{6.4}{-1} = \frac{-\log_{10}[\text{H}^+]}{-1}$$

$$10^{-6.4} = [\text{H}^+]$$

OJ

$$\frac{4}{-1} = \frac{-\log_{10}[\text{H}^+]}{-1}$$

$$10^{-4} = [\text{H}^+]$$

comparison

$$\frac{10^{-4}}{10^{-6.4}} = \boxed{251.18}$$

$\therefore$ OJ is 251.18 more times acidic than milk

(2)

- g) The ratio of intensities of sound pollution measured at a small airport runway versus the local highway is 6420.4. If the sound level on the local highway is 91 dB, determine the sound level on the runway.

$$\beta_2 - \beta_1 = 10 \log_{10}\left(\frac{I_2}{I_1}\right)$$

$$\beta_2 = 10 \log_{10}\left(\frac{I_2}{I_1}\right) + \beta_1$$

$$\beta_2 = 10 \log_{10}(6420.2) + 91$$

$$\beta_2 = \boxed{129.08}$$

the sound on the runway is

129.08 dB

(2)

h) Solve for x given, $\frac{\log(28-x^3)}{\log(4-x)} = 2$.

$$\log_{(4-x)} (28-x^3) = 2$$

$$(4-x)^2 = (28-x^3)$$

$$(4-x)(4-x) = 28-x^3$$

$$16-8x+x^2 = 28-x^3$$

$$x^3+x^2-8x+16-28=0$$

$$x^3+x^2-8x-12=0$$

$$(x-3)(x+2)^2$$

$$x = 3, -2$$

check:

$$\frac{\log(28-(3)^3)}{\log(4-(3))} = 2$$

$$\log(4-(3))$$

↑ having in this term $\rightarrow \log_{(4)}(28-(3)^3)$

↳ not possible

so invalid answer

$$\frac{\log(28-(-2)^3)}{\log(4-(-2))} = 2 \checkmark$$

$\therefore$ the only applicable value of $x = -2$
and $x = 3$ is invalid

(4)

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